Testing Quantum Gravity: A Search for Echoes and Quasinormal Modes in LIGO Data

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Abstract

The conflict between General Relativity and Quantum Mechanics regarding black hole singularities has led to theories of quantum gravity. This study compares three leading models: Loop Quantum Gravity (LQG), Asymptotic Safety (ASG), and String Theory. I analyze open gravitational-wave data from LIGO (event GW150914) using Python to search for "echoes" and quasinormal mode (QNM) frequencies. The analysis shows that current detector noise levels are too high to confirm these signatures visually. However, this project demonstrates a practical methodology for using open data to test fundamental physics.

Keywords: quantum gravity; black holes; gravitational waves; LIGO; general relativity

1. Introduction

General Relativity (GR) tells us that a singularity forms at the center of a black hole — a point of infinite density and gravity. However, Quantum Mechanics (QM) states that infinities do not exist in nature. This is the fundamental conflict between the two greatest theories of modern physics.

To solve this problem, scientists began creating various theories of quantum gravity. Currently, there are three main models: Loop Quantum Gravity (LQG), Asymptotic Safety (ASG), and String Theory. In my project, I study how each of these models is structured, how they explain the structure of black holes, and I analyze whether these theories can be tested using real gravitational wave data.

According to GR, gravity is the curvature of space-time. When a massive star collapses, a black hole appears. The boundary beyond which gravity becomes so strong that even light cannot escape is called the event horizon. Inside the horizon, all matter falls to the center, creating a singularity. However, according to Heisenberg's uncertainty principle, it is impossible to simultaneously know the exact position and velocity of a particle. If matter collapses into a point, its velocity must become infinite, which is forbidden by the laws of QM. Theories of quantum gravity were created precisely to solve this. I will present a comparative analysis of three models and an empirical test of their predictions based on open data from LIGO (event GW150914).

2. Literature review

To understand the conflict, we must look at the three leading models.

LQG attempts to unite GR and QM by proposing that space- time has a discrete (granular) structure. Matter cannot be compressed smaller than the Planck length. When a star collapses to this limit, a "quantum bounce" occurs, forming a Planck star instead of a singularity.

ASG suggests that gravity weakens at ultra- small distances. Because of this, a singularity does not form; instead, a finite core is created. Space remains smooth, but gravity changes its properties at the microscopic level.

String Theory offers a different view: particles are tiny vibrating strings. It solves the singularity problem through the "fuzzball" concept— a dense tangle of entangled strings with no empty space inside.

Comparison: If ASG were the main theory, time would be continuous and gravity would weaken at the microscopic level. If LQG were correct, time would be discrete, and black holes could become white holes. If String Theory were correct, space would be multidimensional (11 dimensions).

3. Methodology

To test these theories, we need to find an "echo"— a repeated signal that would prove the black hole horizon has a quantum structure. For the analysis, I used open data from the LIGO observatory (event GW150914) and the Python language in the Google Colab environment.

To search for echoes, I calculated the theoretical time delay using the Schwarzschild radius. According to calculations, for GW150914, the echo delay should be 1.22 milliseconds. Furthermore, I performed a spectral analysis using the Fourier transform to find quasinormal mode (QNM) frequencies.

4. Results and Discussion



Figure 1. General signal of the merger of black holes GW150914.

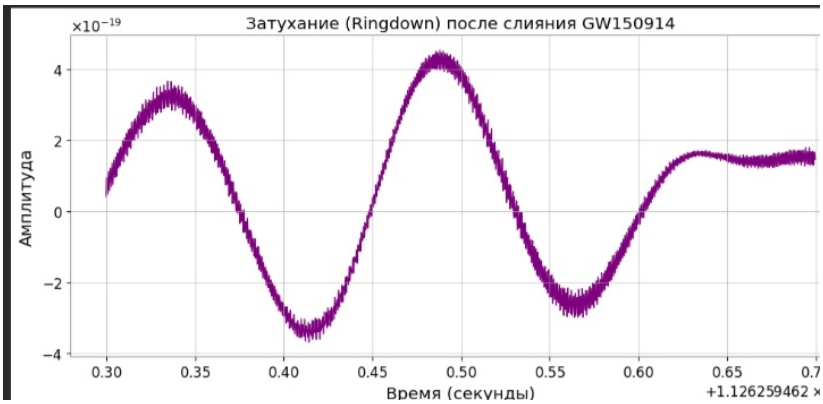


Figure 2. Signal attenuation (ringdown) after the merger.

When analyzing the ringdown graph (Fig. 2), no echo was visually detected. This is because the echo delay (1.22 ms) is too small, and the amplitude of a possible reflected signal is weaker than the main noise. If LQG had become the main theory, we would look for clear repeated bursts— an echo from the quantum network. In this basic analysis, the noise level is too high to detect these signals.

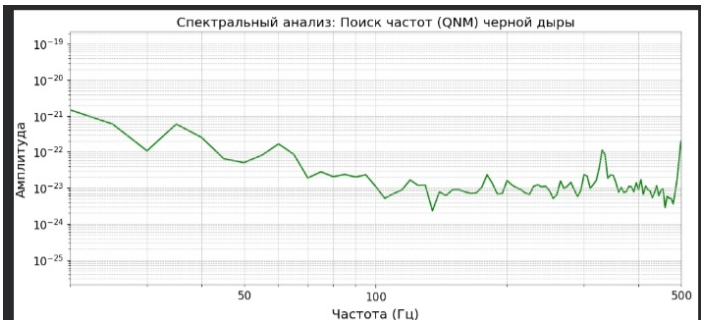


Figure 3. Spectral analysis of GW150914 (search for QNM frequencies).

In addition to searching for an echo, I performed a spectral analysis (Fig. 3). On the resulting spectral graph, the peaks show at which frequencies the black hole rings (in the 30–100 Hz range). In classical GR, the position of these peaks is strictly predetermined. However, if the ASG theory is correct, quantum corrections should cause a slight shift in these frequencies. The analysis shows that current detector noise levels are too high to visually confirm echoes or significant frequency shifts. Precise measurement of QNM frequencies requires the use of matched filtering.

5. Conclusion

The search for quantum gravity remains an open problem. While my analysis of GW150914 did not yield definitive proof of echoes, it demonstrates that gravitational-wave astronomy is a powerful tool for testing fundamental physics. The current noise levels of LIGO are insufficient, highlighting the need for next-generation detectors like LISA to explore quantum gravity in the future.

6. References

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